INTERNAL DEFENSE PRESENTATION

Development of a Secure Web-Based Medical Image Management System Using Steganography

Protecting patient diagnosis data inside medical images without visible distortion

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Presentation Outline

Large-type version for dim projectors and back-row readability

Introduction

Medical image security problem

Medical images carry context

X-rays and scans are usually shared with notes, findings, and patient identifiers.

Steganography hides presence

Sensitive text is embedded inside the image itself, while the image still looks normal.

This project

A secure web prototype for embedding and recovering patient diagnosis data from medical images.

Confidentiality | Image Quality | Audit Logging | Web Access

Review of Literature and Related Works

Objective • Methodology • Result • Contribution

Image steganography studies

Prior work shows the key trade-off: hiding capacity versus image imperceptibility and robustness.

Medical-image quality metrics

Related works use MSE and PSNR to confirm that embedded images remain visually reliable.

Gap addressed

This project combines LSB steganography with authentication, browser workflow, and audit logging.

Motivation

Why the system is needed

- Patient diagnosis notes must remain confidential during image transfer.
- Medical images should not be visibly degraded by the security process.
- Users need a simple web interface rather than command-line tools.
- Operations should be traceable through audit records.

Objectives

Project targets

- Build authenticated access to the medical image system.
- Embed patient diagnosis text into PNG/JPG/DICOM-style image workflows.
- Recover hidden diagnosis text accurately during decryption.
- Preserve visual quality of the steganographic output image.
- Store encryption and decryption actions in an audit log.

Methodology

Least Significant Bit embedding workflow

1. Prepare inputs

Upload the cover medical image and type the patient diagnosis payload.

2. Embed data

Convert diagnosis text to bits and hide them in the least significant RGB pixel bits.

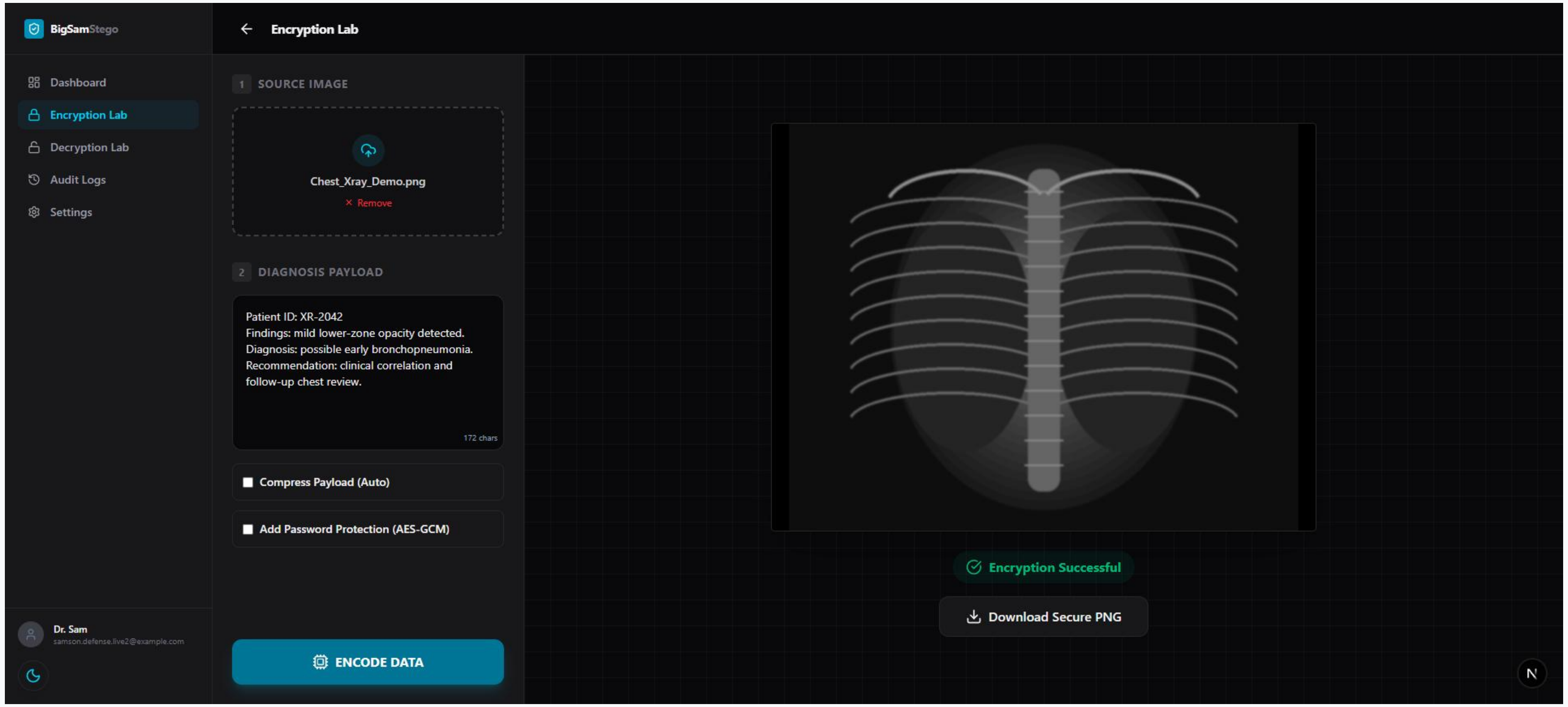
3. Recover & log

Upload the stego-image, extract the hidden text, and record the operation in the audit log.

Cover image | Diagnosis text | LSB algorithm | Stego PNG | Recovered text

Result and Discussion

Live browser screenshot: Encryption Lab



Real test

Uploaded demo X-ray and entered patient diagnosis text.

Output

System produced a downloadable secure PNG.

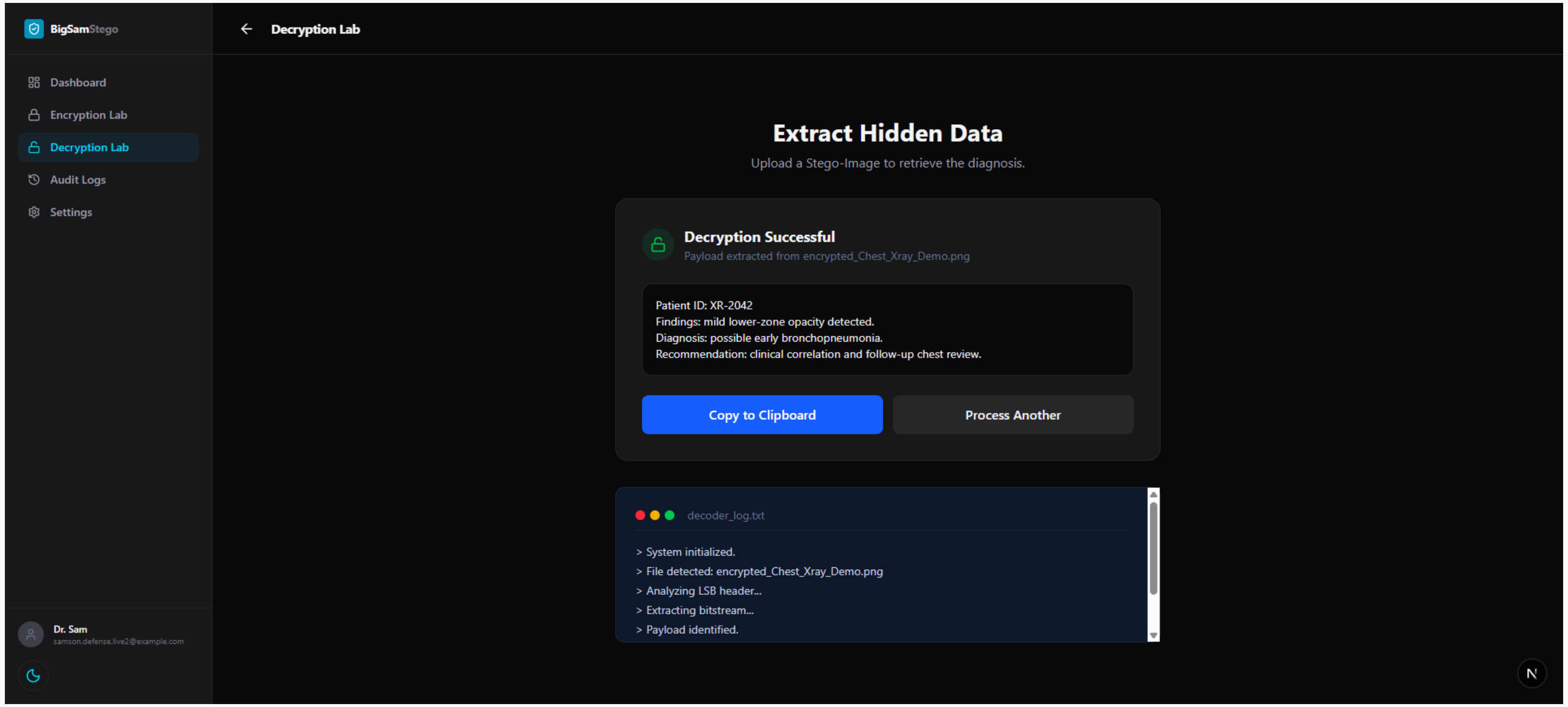
Audit

Encryption event was recorded by the app.

Figure 4.1: Live Encryption Lab screenshot showing uploaded X-ray, diagnosis payload, and success state.

Result and Discussion

Live browser screenshot: Decryption Interface



Input

Uploaded the encrypted PNG generated by the browser test.

Recovered

Diagnosis text was extracted successfully.

Verification

Terminal log shows payload identified and data extracted.

Figure 4.2: Live Decryption Lab screenshot displaying successful recovery of hidden diagnosis data.

Contribution to Knowledge

What the project adds

Practical secure workflow

Pairs medical images with diagnosis text without visible external metadata.

Browser-side prototype

Demonstrates client-side Canvas processing for steganography in a web app.

Authentication + audit

Adds controlled access and activity records to the steganographic process.

Defense-ready evidence

Includes live screenshots, functional testing, and quality discussion.

Conclusion and Recommendation

Summary and future work

Conclusion

The system successfully embeds and retrieves patient diagnosis data from medical images. The live browser test confirmed encryption, decryption, recovery, and audit behavior.

Recommendations

- Test with larger DICOM datasets
- Combine LSB with stronger encryption
- Add role-based clinical access control
- Deploy with PostgreSQL and secure cloud storage

References

Selected sources

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- Provos, N., & Honeyman, P. (2003). Hide and seek: an introduction to steganography.
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- Project materials: Chapter Four.docx, Literature Review PDF, README, and implemented Next.js app.

Thank you